

Chinese-born theoretical physicist Yang Chen-Ning revolutionized the field of particle physics with his groundbreaking research into subatomic decay, for which he received the Nobel Prize in Physics.

After obtaining a PhD in nuclear physics from the University of Chicago, Yang Chen-Ning moved to Princeton to study subatomic decay (the transformation of one subatomic particle into others) in 1949. At the time, a law of physics called conservation of parity stated that it is not possible to distinguish between different directions in particle interactions. Yang's work led him to question this law.

In 1956, after extensive research with partner Tsung-Dao Lee, the pair proposed the law was invalid for certain so-called "weak" interactions, such as radioactive decay. They also suggested an experiment that might prove their proposition was correct. The experiment was successfully carried out by Chien-Shiung Wu (see p.253) later that year, and won Yang and Lee the Nobel Prize in Physics in 1957.

"These **concepts** were **not dreamed up**. They were **natural and real**."

Shiing-Shen Chern, 1975

YANG CHEN-NING

MILESTONES

US SCHOLARSHIP

Leaves China after being awarded a scholarship at the University of Chicago to study physics in 1946.

PARTICLE THEORY

Proposes the Yang-Mills theory on elementary particles with his colleague Robert Mills in 1954.

BREAKTHROUGH PAPER

Publishes paper in 1956 to prove conservation of parity is fallible; awarded Nobel Prize for his work in 1957.

RESEARCH DIRECTOR

Becomes director of the theoretical physics research center at Stony Brook University in 1965.